

SocialSim: An Open Source Platform for Conducting Behavioral Intervention Research on Social Media

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Abstract

Understanding adolescent online behavior is critical for designing effective online safety interventions; yet, studying such behavior ethically and realistically remains a challenge. To address this, we present SocialSim, a social media simulation tool that builds on an open-source platform designed to enable in situ experimentation in a safe, controlled environment. SocialSim replicates key features of social platforms, including feed, profiles, friending, and private messaging while offering a flexible architecture for researchers to study user interactions and deploy behavioral interventions, such as real-time online safety nudges within direct messaging, which are unsupported in other platforms. We used this tool with a Wizard-of-Oz approach, combined with automated, as well as scripted interactions which allows researchers to simulate lifelike social media while maintaining a safe environment for youth. This demonstration presents the design, technical implementation, and future research implications for this simulation as a sandbox for evaluating online safety interventions, and as a generalizable platform for ethically studying human behavior in an ecologically valid social media environment.

CCS Concepts

• Human-centered computing → User interface management systems.

Keywords

Nudges, Online Risks, Social Media, Simulation, Interventions, Ecological Validity

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1 Introduction

Despite growing interest in studying social behaviors online, researchers often face ethical and methodological challenges when conducting in situ experiments that require realism, control, and participant safety [6, 21]. To address this gap, we introduce SocialSim; a simulated platform, that mimics key social media features and caters to the needs of both researchers and participants in providing a safe, customizable and realistic environment for evaluating human behavior within social media. A unique aspect of the simulation is that it facilitates a hybrid of the Wizard-of-Oz method [13], combined with automated interactions, which enables both researcher-mediated and programmatically triggered interventions. Automated scripts can simulate engagement from other social media users on the feed, while researchers can serve as “wizards” using the chat feature for conversational interaction that triggers interventions.

We demonstrate the value of SocialSim through a case study in adolescent online safety research, a domain where prior work has explored design interventions [2, 16] to support teens facing various online risks [8, 9]. Using this simulation, we conducted evaluations with teens through scripted and researcher-mediated low-risk online scenarios, where nudges are triggered in response to contextual cues in both public and private interactions [5], without exposing teens to real harm. By lowering the barrier to creating safe and realistic experimental conditions, this tool offers a flexible platform for studying online interventions in contexts that would otherwise be difficult to safely replicate in the wild, across a range of research domains within HCI and social computing.

1.1 Gaps Within Existing Social Media Evaluation Tools

Most prior evaluations within adolescent online safety employed survey-based feedback [12, 16] which have provided useful insights into online behavior and intervention design. Yet, these approaches often lack contextual nuance and realism [17], where hypothetical responses may diverge from actual behavior, emphasizing the need for ecologically valid evaluations [21] for behavior change [15, 19]. Within the fields of privacy and security, there have been some ecologically valid evaluations of nudges that represent the user’s real-world behaviors [7, 14]. Yet, few tools exist that allow researchers to flexibly simulate interactive social media experiences

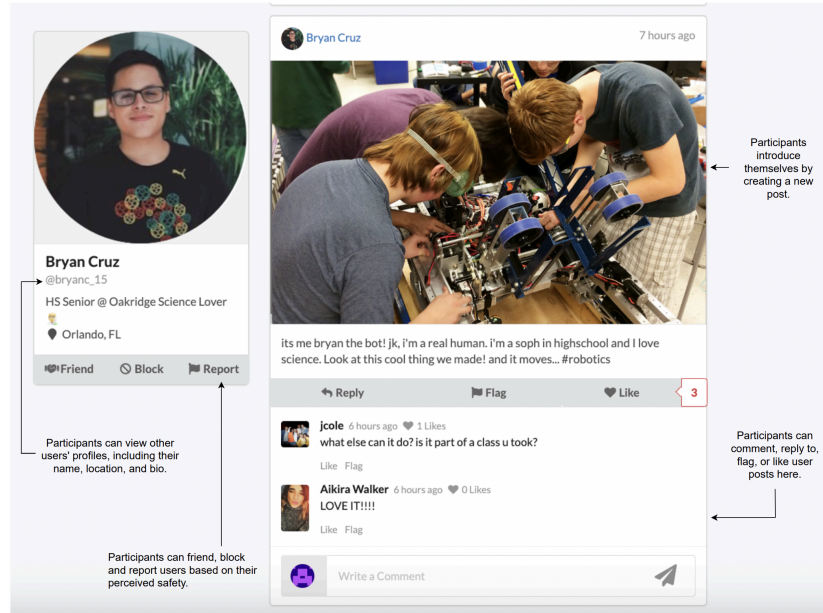


Figure 1: User Profile, Feed and Friending

in a controlled and safe way. Building upon prior works, we conducted meta-level research with teens to understand and co-design effective ways of simulating an environment for research evaluation, reflected in our system [4, 5]. Compared to other social media simulation tools, [1], our platform offers the adaptability of integrating personas, scripts and interventions, that is broadly applicable for researchers to conduct effective evaluations within social media contexts [5].

2 SocialSim: Implementation and Design

The social media simulation builds on the technology of an open-sourced platform called *Truman*, based on The Truman Show [10], developed as a customizable web-based social media site for research experimentation by DiFranzo et al. [11]. Below, we describe both the technical implementation of SocialSim and the core features of this system.

2.1 Technical Implementation

We built our simulation using an open-source, web-based platform using Node.js¹ and MongoDB² on the backend with jQuery and Semantic UI framework on the client side. The platform follows standard security practices, such as HTTPS, Cross-Site Request Forgery protection, and Content Security Policy headers. For further security, we implemented an invite-only system, using one-time access codes to prevent scams or malicious actors. We built upon the Truman platform by developing a chat interface, as majority of online risks faced by teens occur in private settings [20]. The chat was built using socket.IO³, node.JS, Pug, MongoDB Atlas, and AWS

S3 for image-based data⁴. The nudges were triggered by matching the chat messages sent to our stored risky keywords. The user action data for nudges, feed interactions, chats and friending was logged in the MongoDB database. The web application has been hosted on AWS EC2⁵ and we plan to open-source the latest version upon publication.

2.2 Design-based Features

We built upon Truman to implement new direct messaging, friending, blocking and reporting features, as well as teen-centered online safety nudges. Within the context of our Wizard-of-Oz evaluation, the system was used by one teen participant at a time, with researchers acting as other active users in the direct messaging, and prepopulated posts and interactions from actor accounts in the feed. Below, we provide an overview of the key features:

- **Sign Up:** Participants can sign up using a name, username, email, password, and a one-time use access code provided by the research team to ensure security.
- **User Profile:** Once signed in, participants are directed to the profile setup page, where they can add a picture, name, location, and a short bio (Fig. 1). The information is visible to other social media users (i.e., actor researchers) and relied upon for building conversations, shared context, and realism for the chat interaction.
- **Feed:** The feed is prepopulated with posts and automated interactions designed by our research team based on teens' feedback in prior work [4]. Participants can scroll through the feed and interact with the posts on the feed and profiles by either liking, commenting, or flagging, based on their preferences (Fig. 1).

¹<https://nodejs.org/en>

²<https://www.mongodb.com/>

³<https://socket.io/>

⁴<https://aws.amazon.com/s3/>

⁵<https://aws.amazon.com/ec2/>

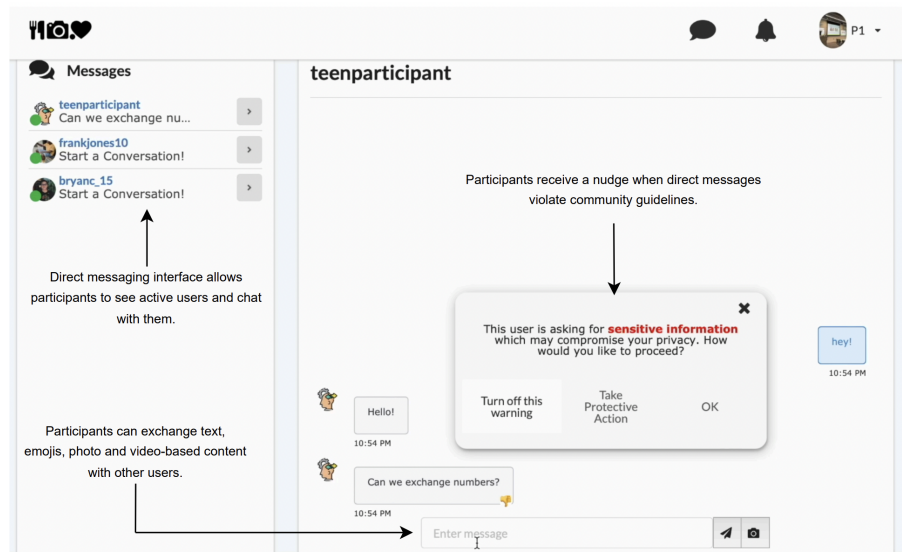


Figure 2: Direct Messaging and Information Breach Nudge

- **Create Posts:** Next, participants create their own posts to introduce themselves to the community. Here, they can add a photo, a caption, and engage with comments on their post.
- **Direct Messaging:** The system offers a real-time private chat interface that supports messaging other users through text, images, and videos (Fig. 2). On the left is a list of users to initiate conversations with. On the right, participants can view unread messages, select to respond and can react to messages.
- **Real-Time Online Safety Nudges:** A central feature of the system is the integration of safety nudges for public and private risks, carefully designed to be triggered by preprogrammed risky comments/messages. The nudges serve as warnings and also provide protective measures to navigate the risky situation safely (Fig. 2).
- **Friending, Blocking and Reporting:** To understand how users perceive safety and form connections online, participants can also friend, block, or report other users from their profiles (Fig. 1).

For our implementation, we integrated six previously co-designed personas as social media users on the system [3, 5], with three risky and three non-risky to balance the environment. One risky persona is an automated cyberbullying comment triggered after participants create a post. The other personas, including two risky ones, are enacted by researchers in private chats who roughly follow predefined scripts, designed to trigger a corresponding safety nudge.

3 Limitations and Future Work

While prior research has explored automated simulations using scripted bot interactions [11], our current system uses a hybrid approach that combines automated components with real-time researcher involvement through the Wizard-of-Oz method. Some built-in features, such as blocking and reporting, function independently, but the implementation of safety nudges in this study relied

on researcher-triggered interactions tied to predefined risk scenarios. This semi-controlled framework allows for contextualized and responsive evaluation, but also highlights the need to move towards "smarter" automated interventions for a variety of personalized risks [2, 18]. One key trade-off in the pursuit of ecological validity is the possible compromise in experimental precision and consistency between participants [6].

Although our implementation focused on the safety of adolescents online, the implications of our system extend beyond this domain. For example, future researchers can adapt the platform to examine how youth interact with AI-driven technologies, such as mimicking chatbots in a controlled and ethically sound manner for mental or sexual health support of youth, rather than exposing them to unpredictable automated agents for sensitive topics. Our broader vision is to support domain-specific research by enabling other researchers to tailor the platform to their own study needs by modifying personas, scripting new nudges, or customizing the simulated actors.

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